Beyond Simple Graphs: Knowledge Graphs

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CS512: Data Mining Principles, 2025 Fall

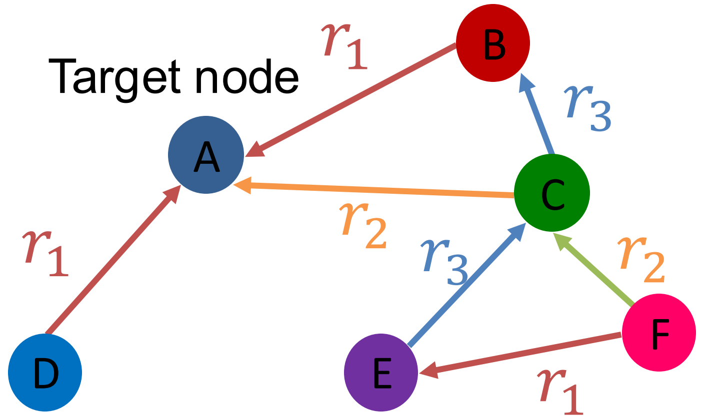
https://ulab-uiuc.github.io/CS512/

Reminder: Coding and Proposal Task

- **Group Idea proposal task** released, due **Oct 15 11:59pm**
  - **Instructions has been released on Canvas**
  - **1.5-2 pages of idea** in ICLR format
  - It will be the proposal for your course project
  - **If you are still unsure about your project, talk to our TA or use their OHs**
- **Coding Assignment 2** released, due **Oct 17 11:59 PM**
  - Submit your code and written answers downloaded from Colab to Canvas

Recap: Heterogeneous Graphs

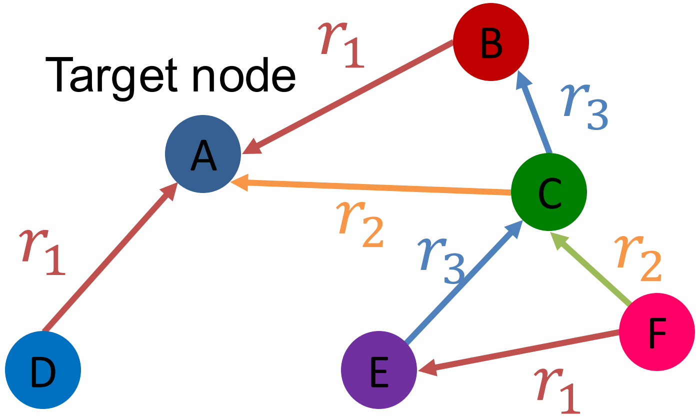
- **Heterogeneous graphs: a graph with multiple relation types**



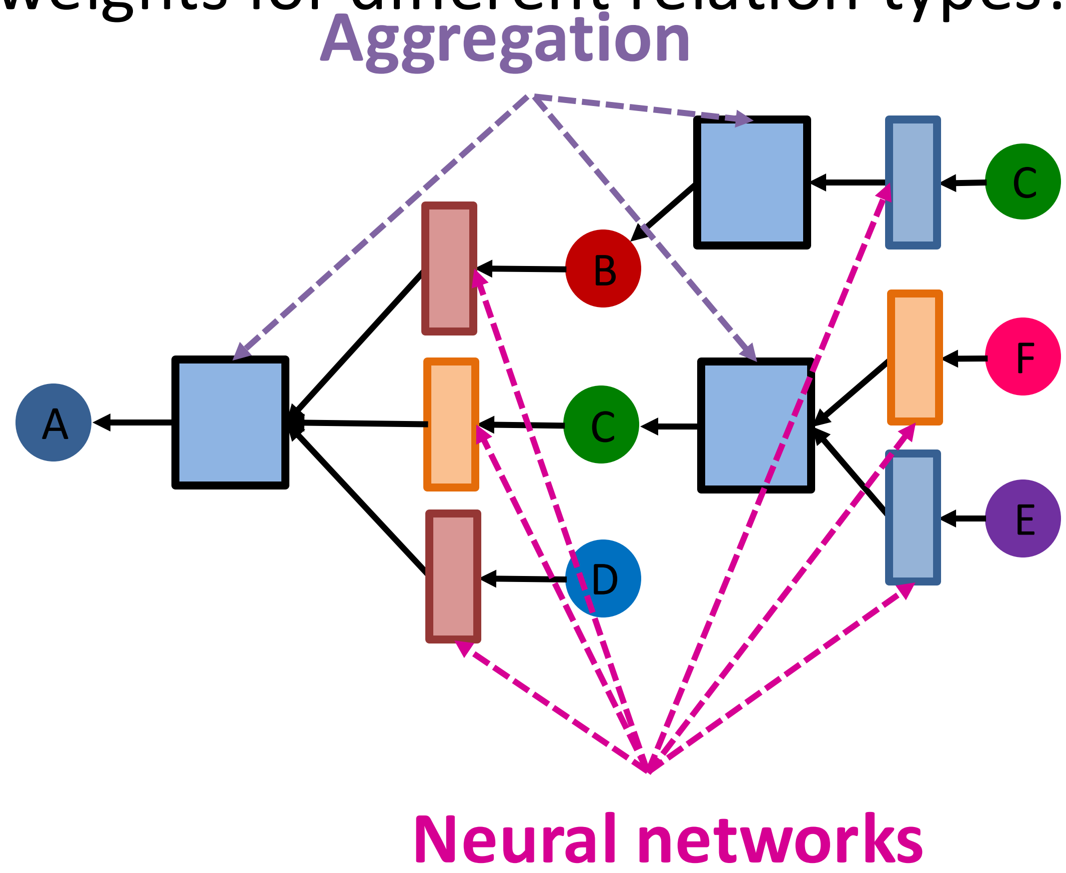
Input graph

Recap: Relational GCN

- Learn from a graph with **multiple relation types**
- Use different neural network weights for different relation types!



Input graph



Today: Knowledge Graphs (KG)

Knowledge in graph form:

- Capture entities, types, and relationships

- Nodes are **entities**
- Nodes are labeled with

Unique ID– **node types**

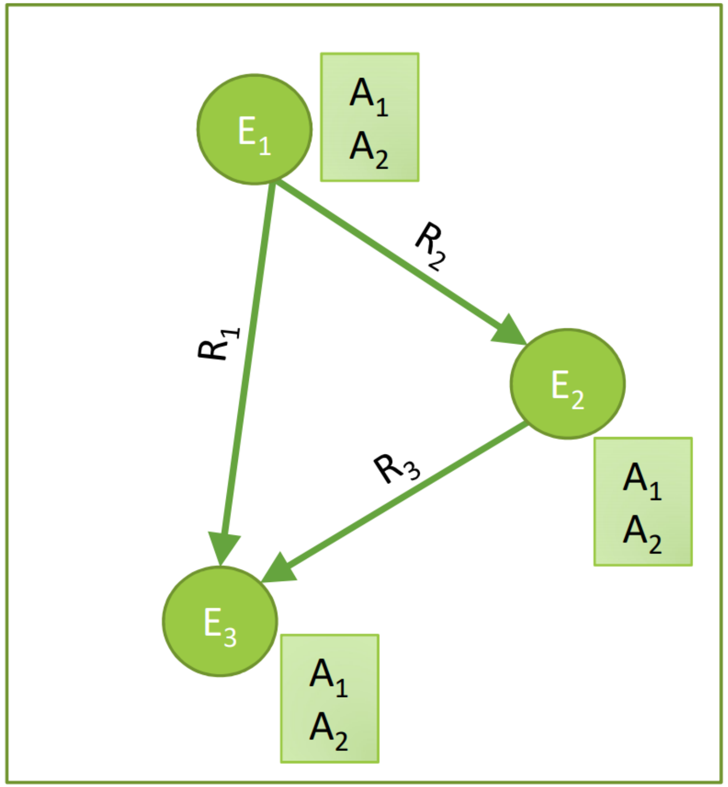
- Edges between two nodes

capture **relationships**

between entities

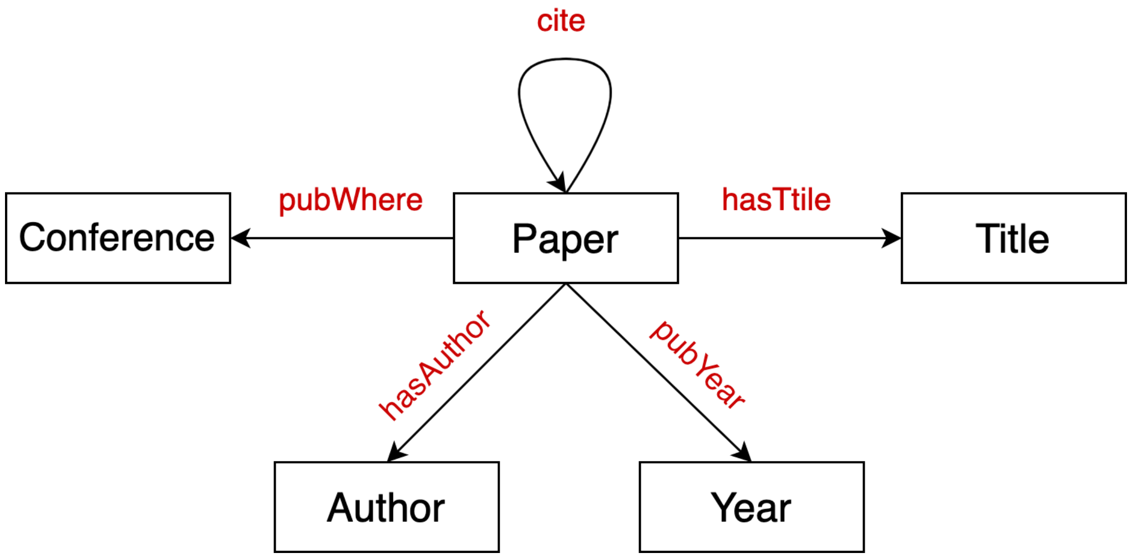
- **KG is an example of a**

heterogeneous graph: each node has a unique type



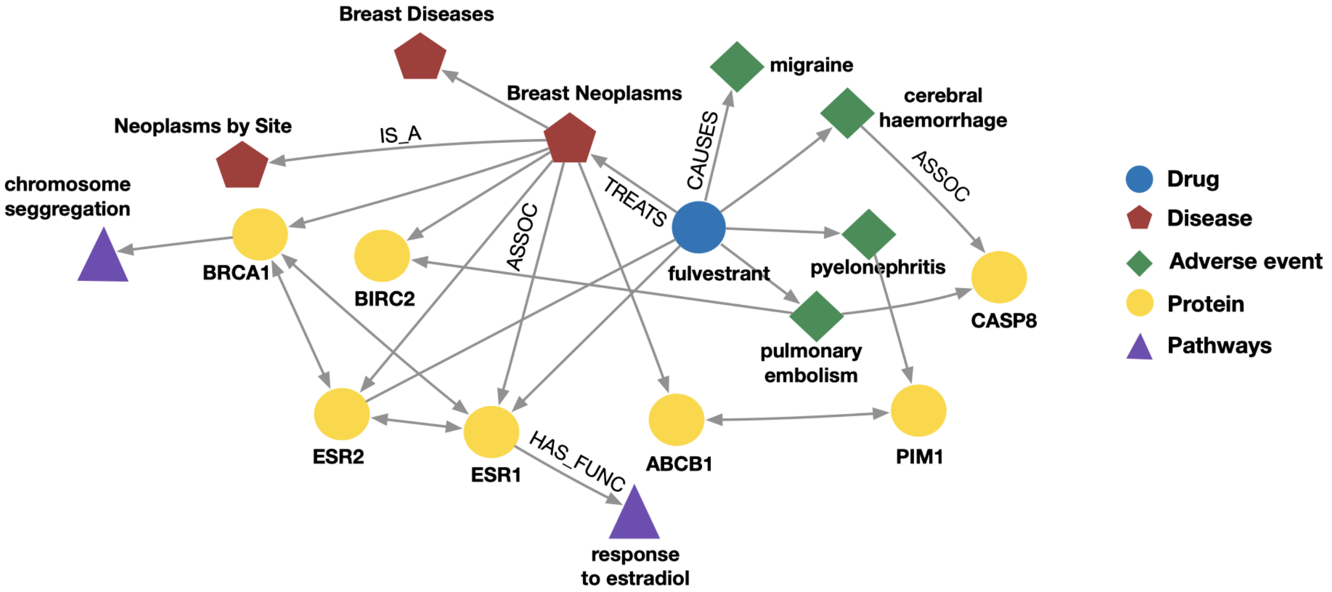
Example: Bibliographic Networks

- **Node types**: paper, title, author, conference, year
- **Relation types**: pubWhere, pubYear, hasTitle, hasAuthor, cite



Example: Bio Knowledge Graphs

- **Node types**: drug, disease, adverse event, protein, pathways
- **Relation types**: has_func, causes, assoc, treats, is_a



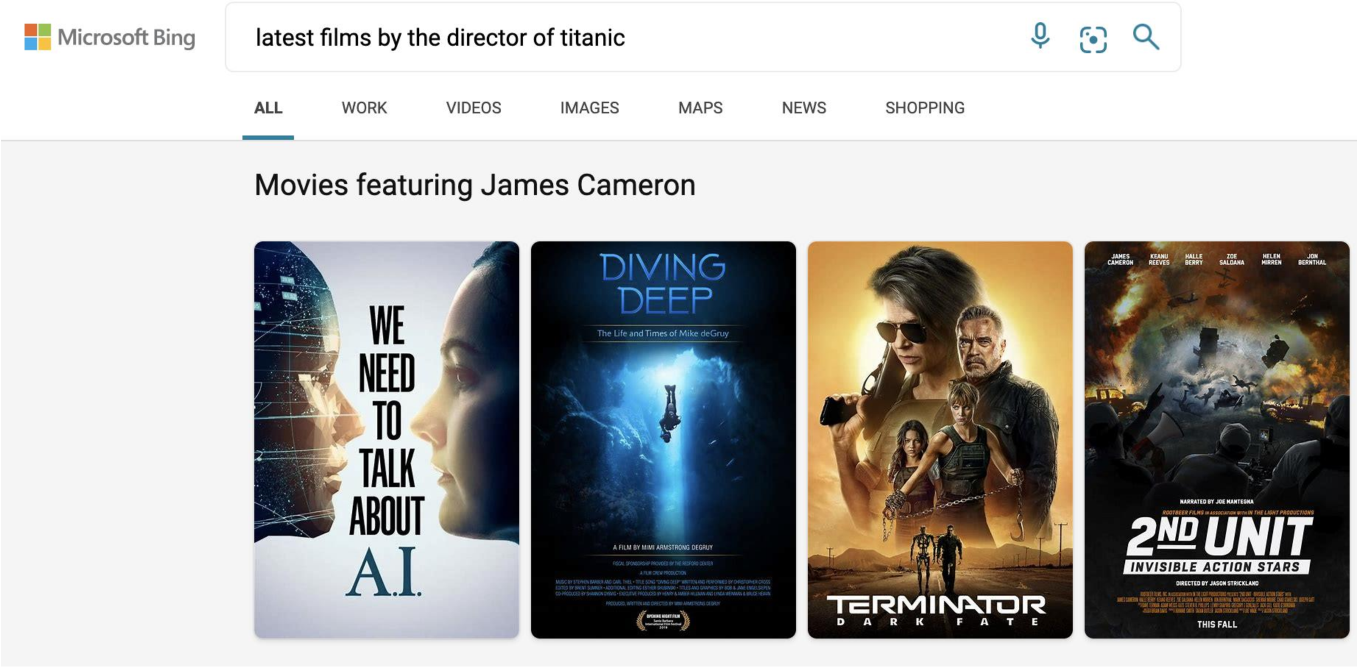
Knowledge Graphs in Practice

Examples of knowledge graphs

- Google Knowledge Graph
- Amazon Product Graph
- Facebook Graph API
- IBM Watson
- Microsoft Satori
- Project Hanover/Literome
- LinkedIn Knowledge Graph
- Yandex Object Answer

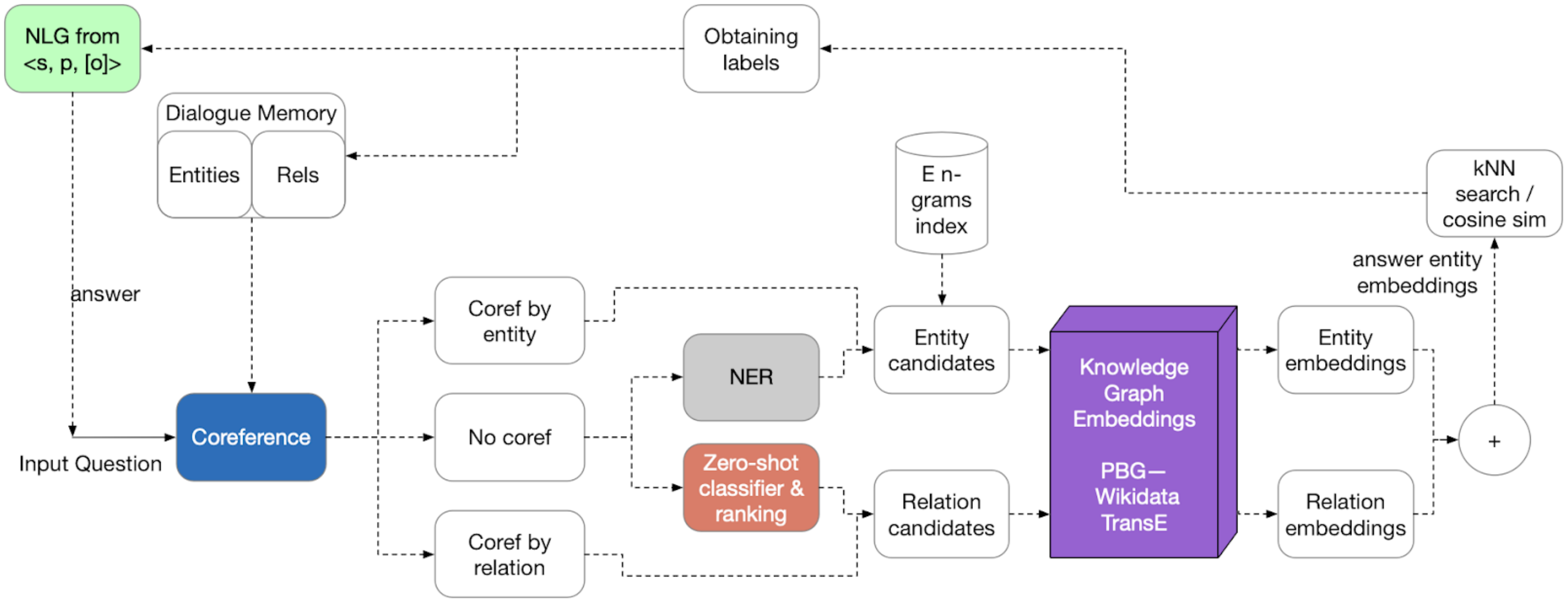
Applications of Knowledge Graphs

- **Serving information:**



Applications of Knowledge Graphs

- **Question answering and conversation agents – the classic approach**



Knowledge Graph Datasets

- **Publicly available KGs:**
  - FreeBase, Wikidata, Dbpedia, YAGO, NELL, etc.

- **Common characteristics:**
  - **Massive**: Millions of nodes and edges
  - **Incomplete**: Many true edges are missing

Given a massive KG, enumerating all the possible facts is intractable! ⇨ **Can we predict plausible BUT missing links?**

Example: Freebase

- **Freebase**
  - ~80 million **entities**
  - ~38K **relation types**
  - ~3 billion **facts/triples**

93.8% of persons from Freebase have no place of birth and 78.5% have no nationality!

- **Datasets:** FB15k/FB15k-237
  - A **complete** subset of Freebase, used by researchers to learn KG models

Dataset	Entities	Relations	Total Edges
FB15k	14,951	1,345	592,213
FB15k-237	14,505	237	310,079

Beyond Simple Graphs: Knowledge Graphs

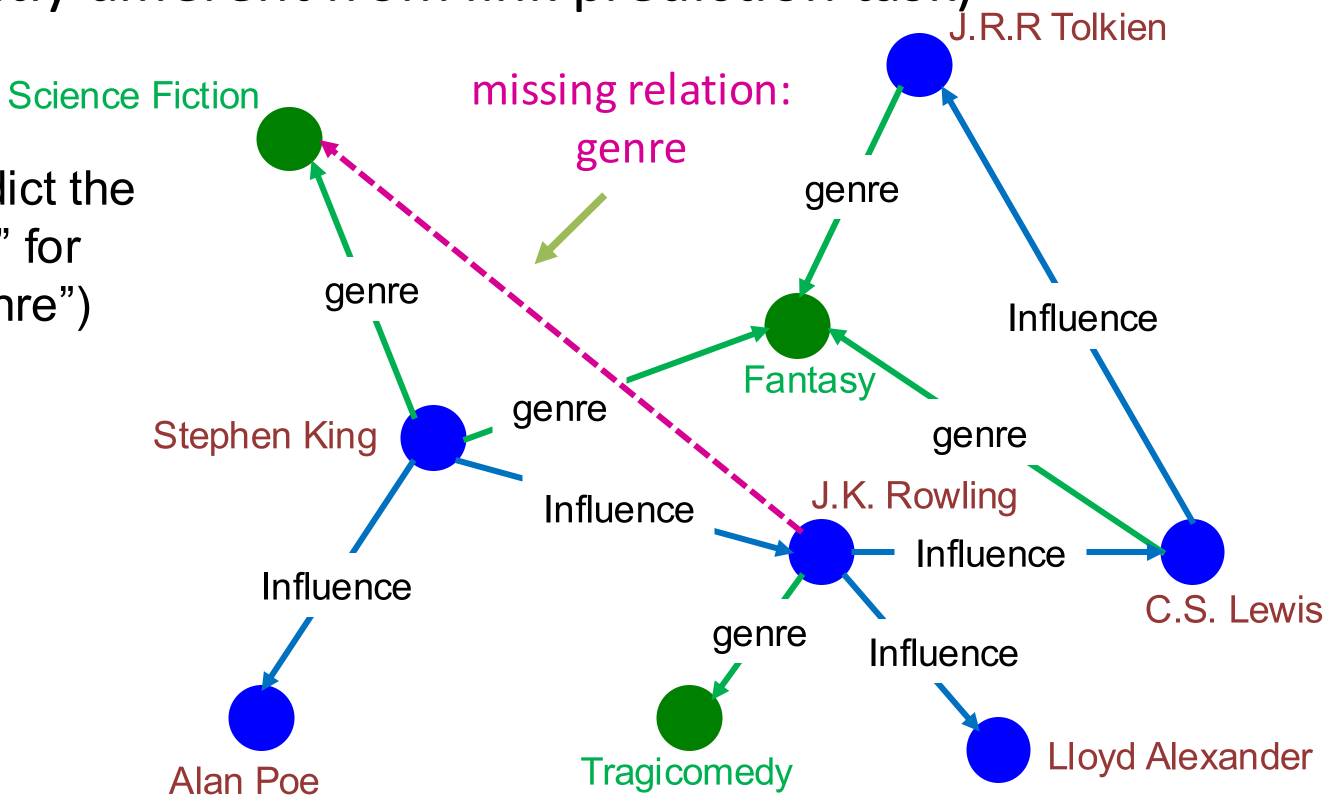
Knowledge Graph Completion

KG Completion Task

Given an enormous KG, can we complete the KG?

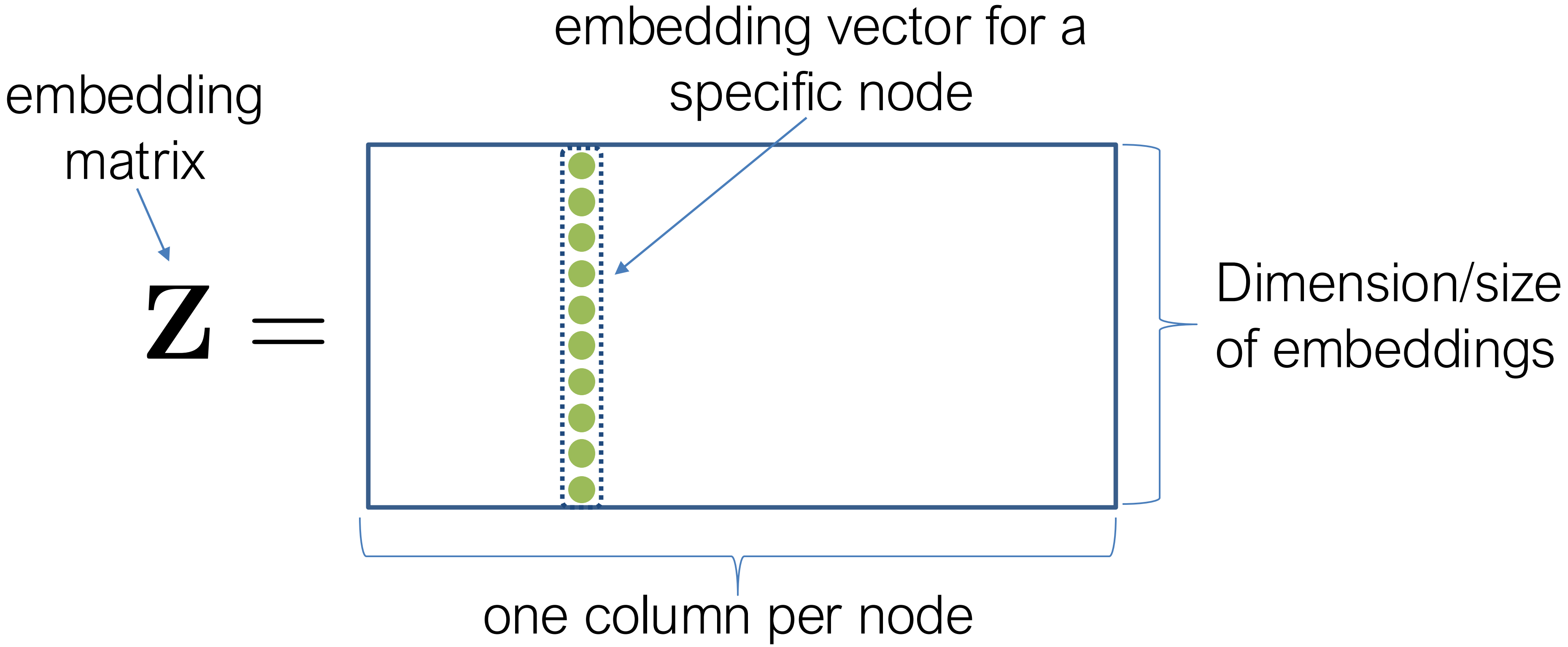
- For a given (**head**, **relation**), we predict missing **tails**.
  - (Note this is slightly different from link prediction task)

Example task: predict the tail "Science Fiction" for ("J.K. Rowling", "genre")



Recap: “Shallow” Encoding

- Simplest encoding approach: **encoder is just an embedding-lookup**



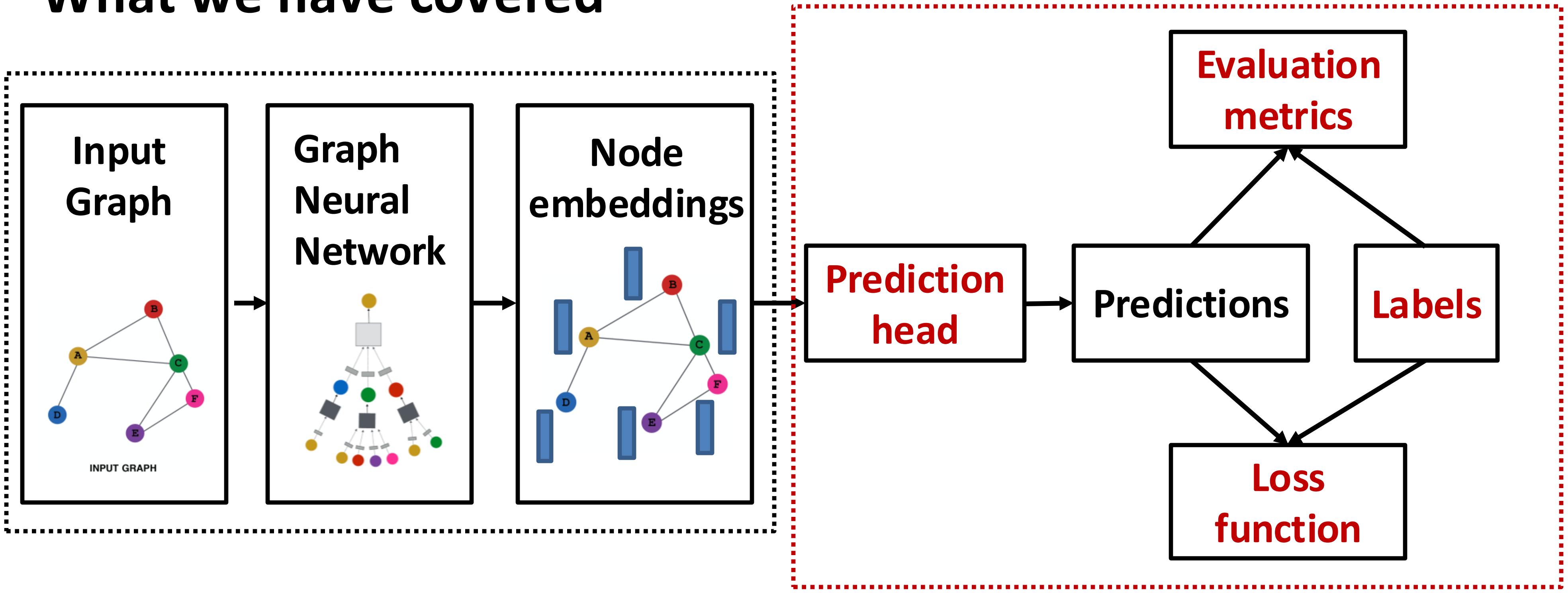
KG Representation

- Edges in KG are represented as **triples** (h, r, t)
  - head (h) has relation (r) with tail (t)
- **Key Idea:**
  - Model entities and relations in embedding space $\mathbb{R}^d$
    - Associate entities and relations with **shallow embeddings (not GNNs)**
      - **Each node and each type of relation has a unique trainable embedding**
  - Given a triple (h, r, t), the goal is that the embedding of (h, r) **should be close** to the embedding of t.
    - How to embed (h, r)?
    - How to define score function $f_r(h, t)$?
      - Score f_r is high if (h, r, t) exists, else f_r is low

KG & GNN Training Pipeline

What KG research is mainly about

What we have covered



Output of a GNN: set of node embeddings

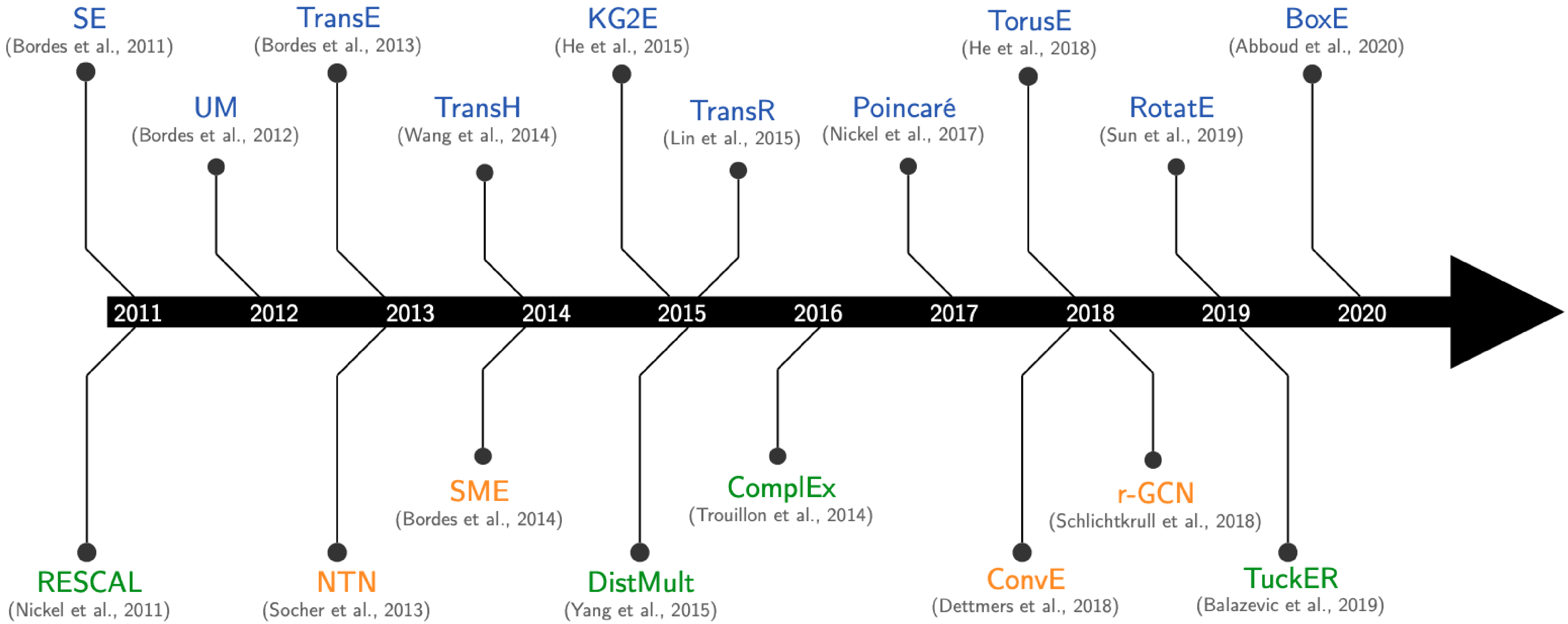
$$\{\mathbf{h}_v^{(L)}, \forall v \in G\}$$

Discussion: How KG Methods Relate to GNNs?

- In essence, **KG methods** are **loss/score functions** defined over **node** and **edge** embeddings
  - Either shallow encoder, or GNNs, can be used to get the embeddings
- Since KGs are heterogeneous graphs with numerous node/relation types, and that there are **no new node/relation types** during test time, **shallow embeddings will be sufficient** in most cases.
  - Recall that the main limitation of shallow embedding is its lack of inductive capability
- **Shallow embeddings** are used to obtain node/edge embeddings for simplicity, but **more advanced deep encoders, e.g., (heterogeneous) GNNs, can be used**

Many KG Embedding

- Many KG embedding Models:



Today: Different Models

We are going to learn about different KG embedding models (shallow/transductive embs):

- Different models are…
  - …based on different geometric intuitions
  - …capture different types of relations (have different expressivity)

Focus today

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓

Beyond Simple Graphs: Knowledge Graphs

Knowledge Graph Completion: TransE

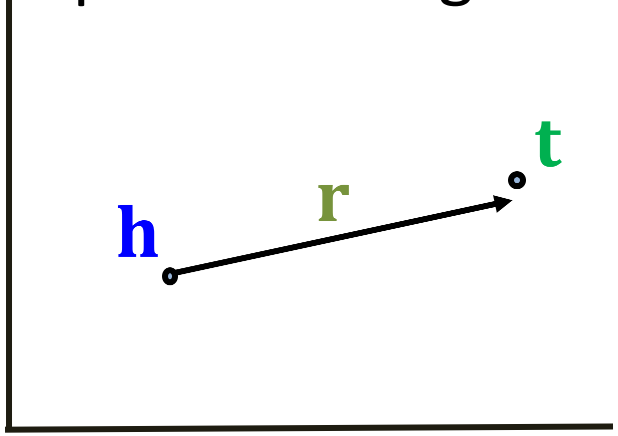
TransE

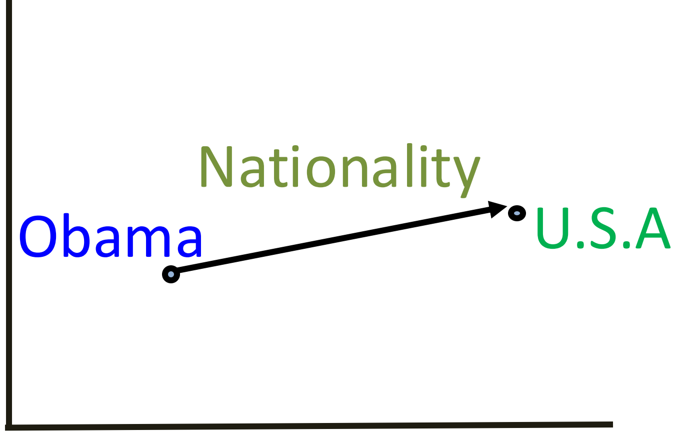
- **Intuition: Translation**
  For a triplet (h, r, t), let $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^d$ be embedding vectors.
- **TransE:** $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ if the given link exists else $\mathbf{h} + \mathbf{r} \neq \mathbf{t}$

Entity scoring function: $f_r(h, t) = -||\mathbf{h} + \mathbf{r} - \mathbf{t}||$

- A valid triplet has a higher score / lower distance

embedding vectors will appear in boldface





TransE: Contrastive/Triplet Loss

Algorithm 1 Learning TransE

input Training set $S = \{(h, \ell, t)\}$, entities and rel. sets E and L, margin γ, embeddings dim. k.

```
initialize ℓ ← uniform(−6/√k, 6/√k) for each ℓ ∈ L
           ℓ ← ℓ/ ‖ℓ‖ for each ℓ ∈ L
           e ← uniform(−6/√k, 6/√k) for each entity e ∈ E
loop
    e ← e/ ‖e‖ for each entity e ∈ E
    S_batch ← sample(S, b) // sample a minibatch of size b
    T_batch ← ∅ // initialize the set of pairs of triplets
    for (h, ℓ, t) ∈ S_batch do
        (h′, ℓ, t′) ← sample(S′_(h,ℓ,t)) // sample a corrupted triplet
        T_batch ← T_batch ∪ {((h, ℓ, t), (h′, ℓ, t′))}
    end for
    Update embeddings w.r.t.
end loop
```

$$\sum_{\left((h,\ell,t),(h',\ell,t')\right) \in T_{batch}} \nabla\left[\gamma + \underbrace{d(\boldsymbol{h} + \boldsymbol{\ell}, \boldsymbol{t})}_{\text{positive sample}} - \underbrace{d(\boldsymbol{h'} + \boldsymbol{\ell}, \boldsymbol{t'})}_{\text{negative sample}}\right]_+$$

Initialize entities ℓ and relations e uniformly, then normalize. γ is margin.

Sample triplet (h', ℓ, t) that does not appear in the KG.

d represents distance (negative of score)

Contrastive loss: Favors lower distance (or higher score) for valid triplets, high distance (or lower score) for corrupted ones

Connectivity Patterns in KG

- **Relations in a heterogeneous KG have different properties:**
  - Example:
    - **Symmetry:** If the edge $(h, \text{"Roommate"}, t)$ exists in KG, then the edge $(t, \text{"Roommate"}, h)$ should also exist.
    - **Inverse relation:** If the edge $(h, \text{"Advisor"}, t)$ exists in KG, then the edge $(t, \text{"Advisee"}, h)$ should also exist.
- Can we **categorize** these relation patterns?
- **Are KG embedding methods (e.g., TransE) expressive enough to model these patterns?**

Four Relation Patterns

- **Symmetric (Antisymmetric) Relations:**

$$r(h,t) \Rightarrow r(t,h) \;\; (r(h,t) \Rightarrow \neg r(t,h)) \quad \forall h,t$$

  - **Example**:
    - Symmetric: Family, Roommate
    - Antisymmetric: Hypernym
- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **Composition (Transitive) Relations:**

$$r_1(x,y) \wedge r_2(y,z) \Rightarrow r_3(x,z) \quad \forall x,y,z$$

  - **Example**: My mother's husband is my father.
- **1-to-N relations:**

$$r(h,t_1), r(h,t_2), \ldots, r(h,t_n) \text{ are all True.}$$

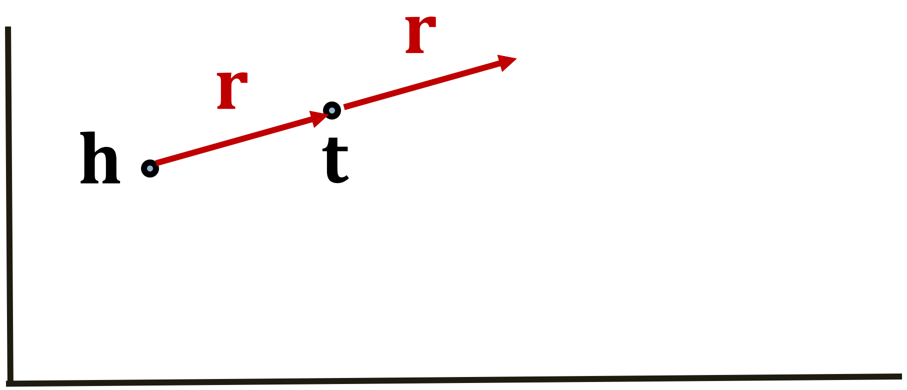
  - **Example**: r is "StudentsOf"

Antisymmetric Relations in TransE

- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

  - **Example**: Hypernym
- **TransE** can model antisymmetric relations ✓
  - $\mathbf{h}+\mathbf{r}=\mathbf{t}$, but $\mathbf{t}+\mathbf{r} \neq \mathbf{h}$

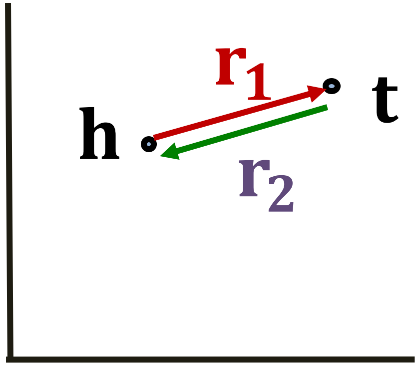


Inverse Relations in TransE

- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **TransE** can model inverse relations ✓
  - $\mathbf{h} + \mathbf{r_2} = \mathbf{t}$, we can set $\mathbf{r}_1 = -\mathbf{r}_2$



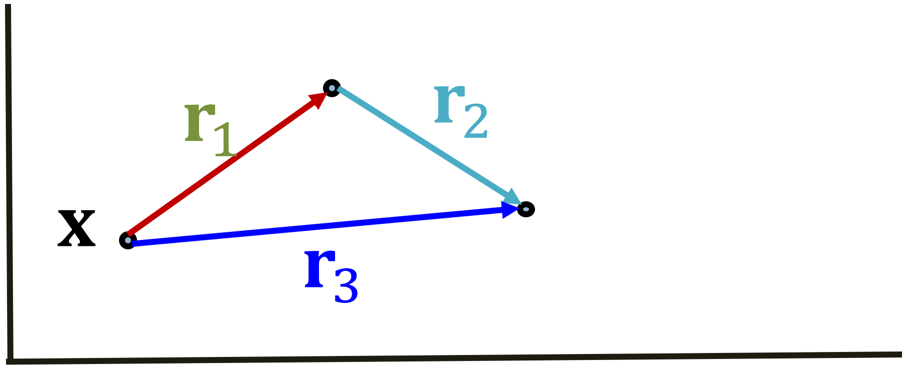
Composition in TransE

- **Composition (Transitive) Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.
- **TransE** can model composition relations ✓

$$\mathbf{r}_3 = \mathbf{r}_1 + \mathbf{r}_2$$



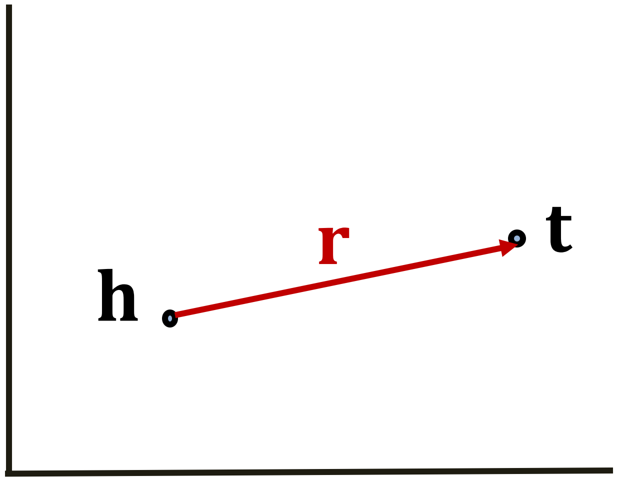
Limitation: Symmetric Relations

- **Symmetric Relations:**

$$r(h, t) \Rightarrow r(t, h) \quad \forall h, t$$

  - **Example**: Family, Roommate
- **TransE cannot** model symmetric relations ✘

  only if $\mathbf{r} = 0, \ \mathbf{h} = \mathbf{t}$



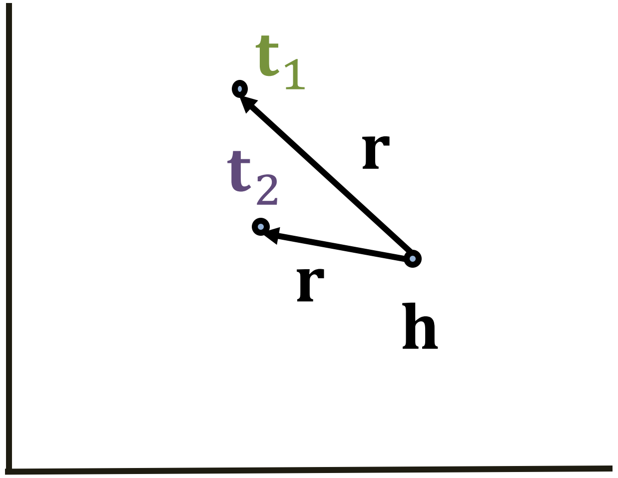
For all h, t that satisfy $r(h, t)$, $r(t, h)$ is also True, which means $||\mathbf{h} + \mathbf{r} - \mathbf{t}|| = 0$ and $||\mathbf{t} + \mathbf{r} - \mathbf{h}|| = 0$. Then $\mathbf{r} = 0$ and $\mathbf{h} = \mathbf{t}$, however h and t are two different entities and should be mapped to different locations.

Limitation: 1-to-N Relations

- **1-to-N Relations:**
  - **Example**: (h, r, t_1) and (h, r, t_2) both exist in the knowledge graph, e.g., r is "StudentsOf"
- **TransE cannot** model 1-to-N relations ✘
  - $\mathbf{t}_1$ and $\mathbf{t}_2$ will map to the same vector, although they are different entities

- $\mathbf{t}_1 = \mathbf{h} + \mathbf{r} = \mathbf{t}_2$
- $\mathbf{t}_1 \neq \mathbf{t}_2$

contradictory!



Today: KG Completion Models

- **What we learned so far:**

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|$	$\mathbf{h},\mathbf{t},\mathbf{r} \in \mathbb{R}^k$	✗	✓	✓	✓	✗

Beyond Simple Graphs: Knowledge Graphs

Knowledge Graph Completion: TransR

TransR

- **TransE models translation of any relation in the same embedding space.**

- **Can we design a new space for each relation and do translation in relation-specific space?**

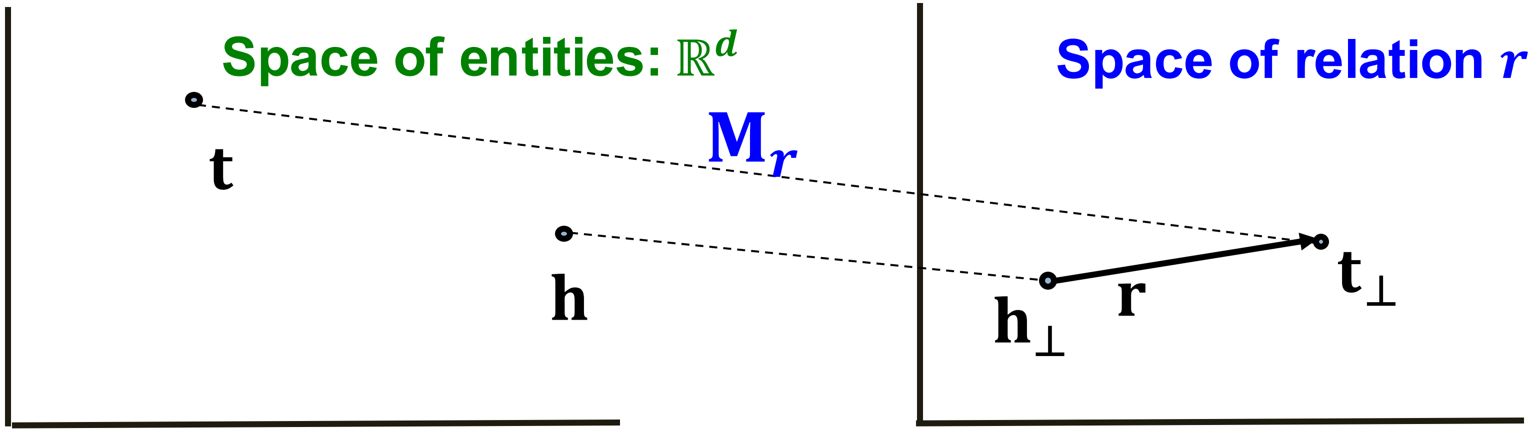
- **TransR**: model **entities** as vectors in the entity space $\mathbb{R}^d$ and model each **relation** as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the projection matrix.

TransR

- **TransR**: model **entities** as vectors in the entity space $\mathbb{R}^d$ and model each **relation** as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the **projection matrix**.

- $\mathbf{h}_\perp = \mathbf{M}_r \mathbf{h}, \;\; \mathbf{t}_\perp = \mathbf{M}_r \mathbf{t}$
- **Score function**: $f_r(h,t) = -||\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp||$

Use $\mathbf{M}_r$ to project from entity space $\mathbb{R}^d$ to relation space $\mathbb{R}^k$!



Symmetric Relations in TransR

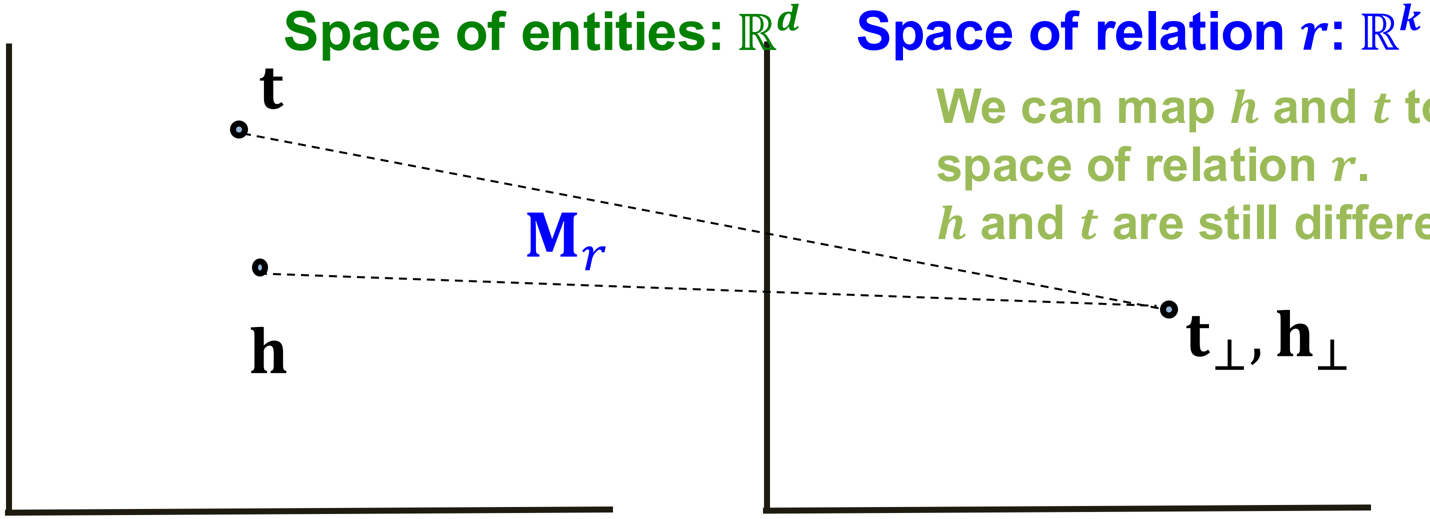
- **Symmetric Relations:**

$$r(h, t) \Rightarrow r(t, h) \quad \forall h, t$$

  - **Example**: Family, Roommate
- **TransR** can model symmetric relations

$$\mathbf{r} = 0, \ \mathbf{h}_{\perp} = \mathbf{M}_r \mathbf{h} = \mathbf{M}_r \mathbf{t} = \mathbf{t}_{\perp} \checkmark$$

Note different symmetric relations may have different $\mathbf{M}_r$



We can map h and t to the same location on the space of relation r.
h and t are still different in the entity space.

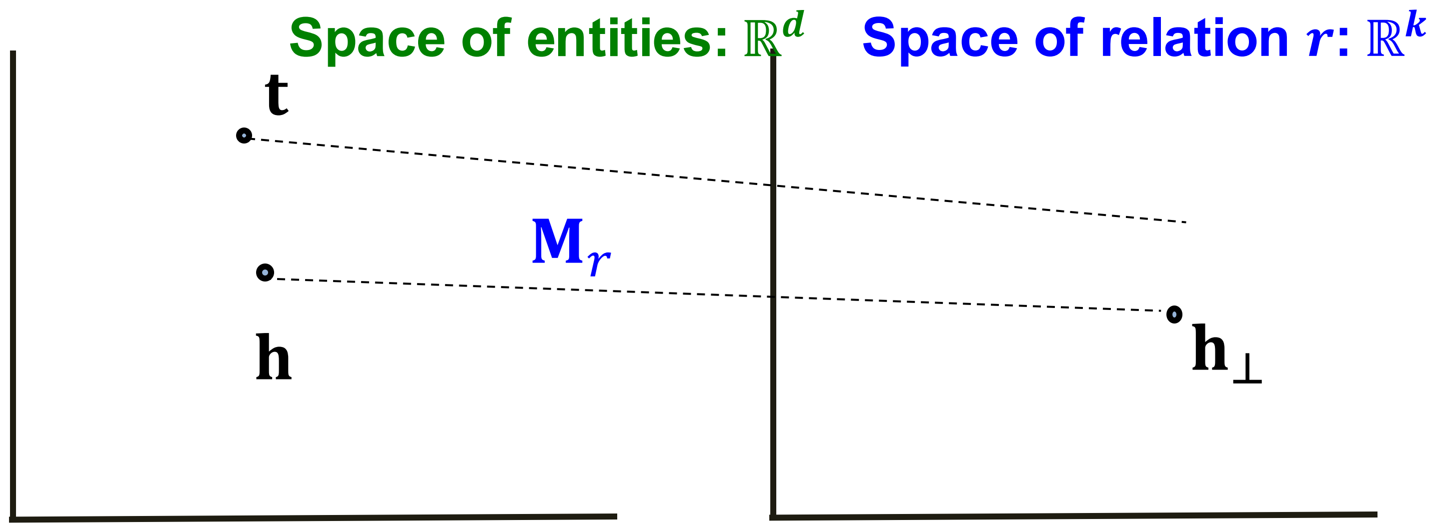
Antisymmetric Relations in TransR

- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

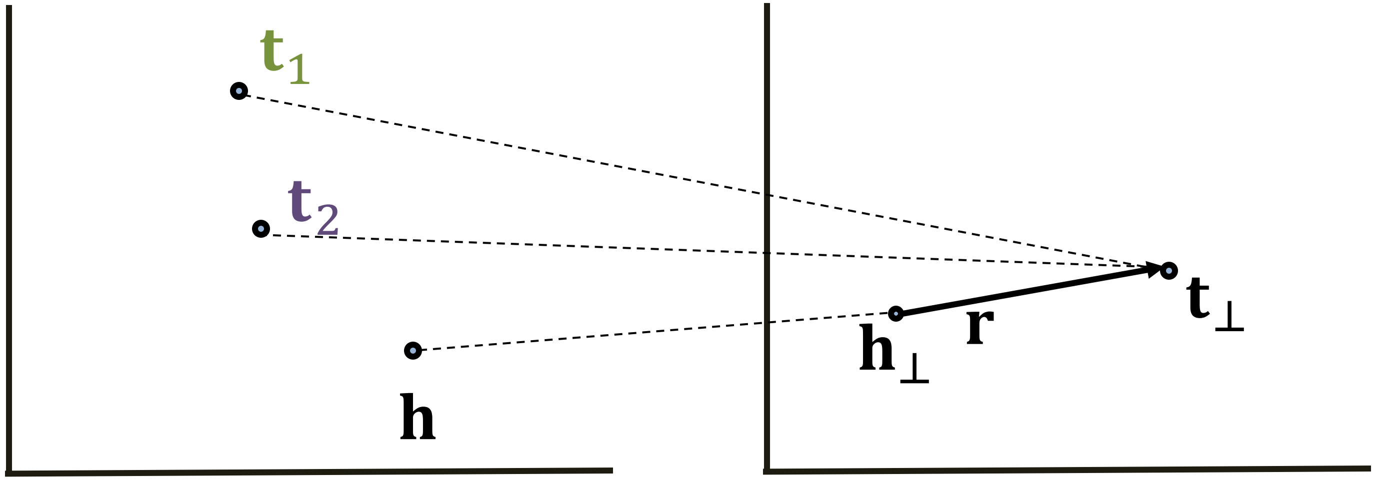
  - **Example**: Hypernym
- **TransR** can model antisymmetric relations:

$$\mathbf{r} \neq 0, \mathbf{M}_r\mathbf{h} + \mathbf{r} = \mathbf{M}_r\mathbf{t}, \text{ Then } \mathbf{M}_r\mathbf{t} + \mathbf{r} \neq \mathbf{M}_r\mathbf{h} \checkmark$$



1-to-N Relations in TransR

- **1-to-N Relations:**
  - **Example**: If (h, r, t_1) and (h, r, t_2) exist in the knowledge graph.
- **TransR** can model 1-to-N relations ✓
  - We can learn $\mathbf{M}_r$ so that $\mathbf{t}_\perp = \mathbf{M}_r \mathbf{t}_1 = \mathbf{M}_r \mathbf{t}_2$
  - Note that $\mathbf{t}_1$ does not need to be equal to $\mathbf{t}_2$!



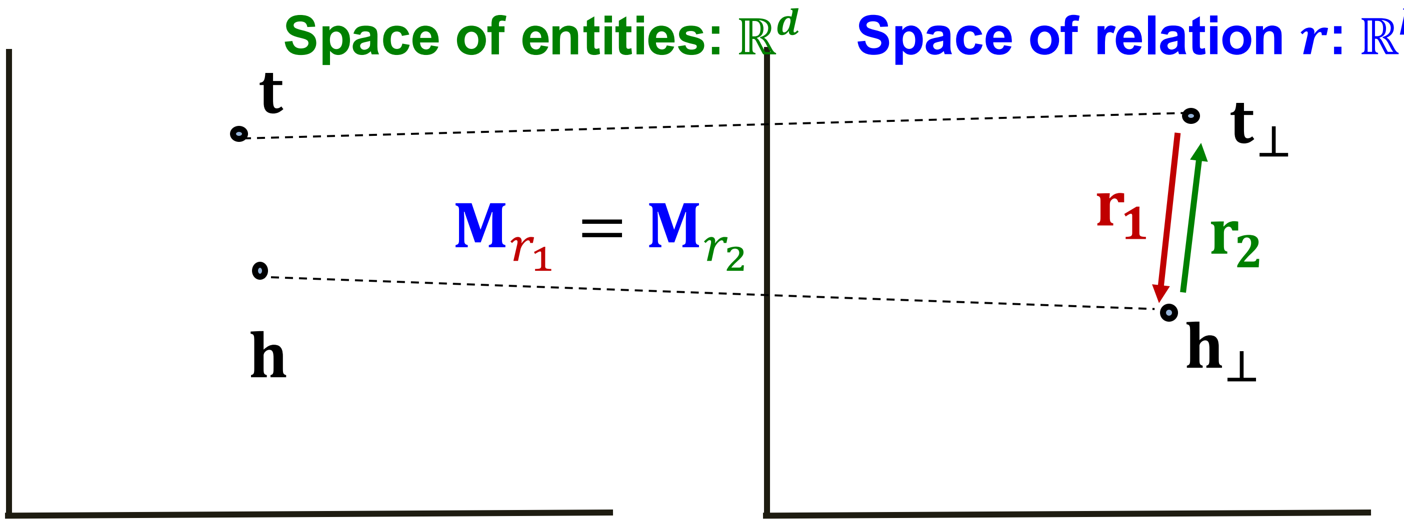
Inverse Relations in TransR

- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **TransR** can model inverse relations

$\mathbf{r}_2 = -\mathbf{r}_1, \mathbf{M}_{r_1} = \mathbf{M}_{r_2}$, Then $\mathbf{M}_{r_1}\mathbf{t} + \mathbf{r}_1 = \mathbf{M}_{r_1}\mathbf{h}$ and $\mathbf{M}_{r_2}\mathbf{h} + \mathbf{r}_2 = \mathbf{M}_{r_2}\mathbf{t}$ ✓



Composition Relations in TransR

- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.
- **TransR** can model composition relations

High-level intuition: TransR models a triplet with linear functions, they are chainable.

Composition Relations in TransR

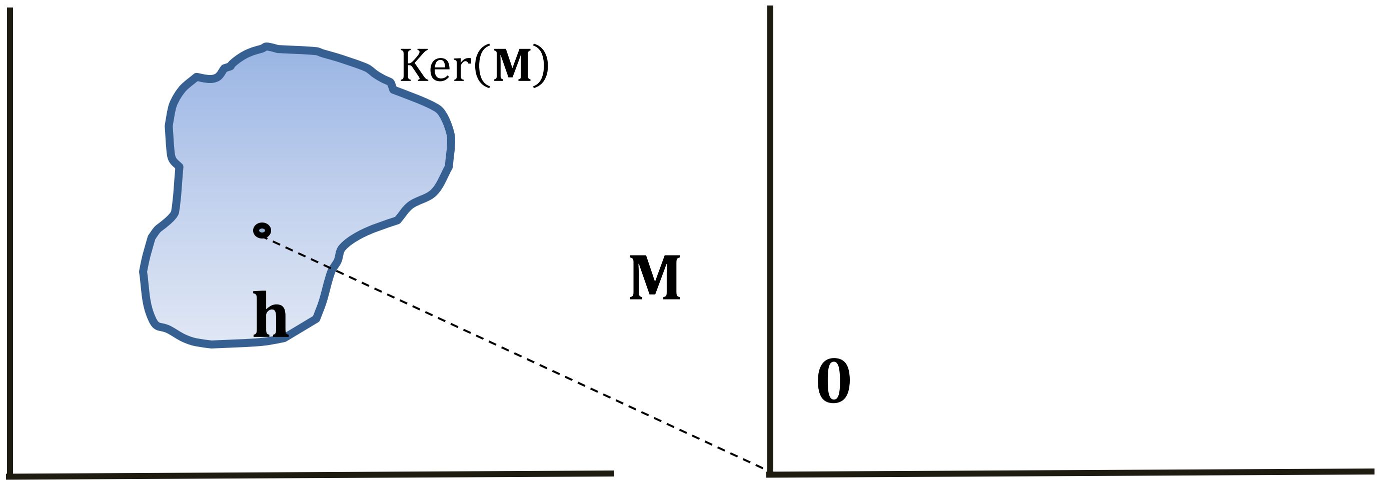
- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

Background:

Kernel space of a matrix $\mathbf{M}$:

$$\mathbf{h} \in \mathrm{Ker}(\mathbf{M}), \text{ then } \mathbf{M} \cdot \mathbf{h} = \mathbf{0}$$



Composition Relations in TransR

- **Composition Relations:**

$$r_1(x,y) \wedge r_2(y,z) \Rightarrow r_3(x,z) \quad \forall x,y,z$$

Assume $\mathbf{M}_{r_1}\mathbf{g}_1 = \mathbf{r}_1$ and $\mathbf{M}_{r_2}\mathbf{g}_2 = \mathbf{r}_2$

- For $r_1(x,y)$:

$r_1(x,y)$ exists $\Rightarrow \mathbf{M}_{r_1}\mathbf{x} + \mathbf{r_1} = \mathbf{M}_{r_1}\mathbf{y} \Rightarrow$

$$\mathbf{y} - \mathbf{x} \in \mathbf{g}_1 + \mathrm{Ker}\left(\mathbf{M}_{r_1}\right) \Rightarrow \mathbf{y} \in \mathbf{x} + \mathbf{g_1} + \mathrm{Ker}\left(\mathbf{M}_{r_1}\right)$$

- Same for $r_2(y,z)$:

$r_2(y,z)$ exists$\Rightarrow \mathbf{M}_{r_2}\mathbf{y} + \mathbf{r_2} = \mathbf{M}_{r_2}\mathbf{z} \Rightarrow$

$$\mathbf{z} - \mathbf{y} \in \mathbf{g}_2 + \mathrm{Ker}\left(\mathbf{M}_{r_2}\right) \Rightarrow \mathbf{z} \in \mathbf{y} + \mathbf{g_2} + \mathrm{Ker}\left(\mathbf{M}_{r_2}\right)$$

- Then, we have

$$\mathbf{z} \in \mathbf{x} + \mathbf{g_1} + \mathbf{g_2} + \mathrm{Ker}\left(\mathbf{M}_{r_1}\right) + \mathrm{Ker}\left(\mathbf{M}_{r_2}\right)$$

Composition Relations in TransR

- **Composition Relations:**

$$r_1(x,y) \wedge r_2(y,z) \Rightarrow r_3(x,z) \quad \forall x,y,z$$

We have $\mathbf{z} \in \mathbf{x} + \mathbf{g_1} + \mathbf{g_2} + \mathrm{Ker}(\mathbf{M}_{r_1}) + \mathrm{Ker}(\mathbf{M}_{r_2})$

- Construct $\mathbf{M}_{r_3}$, s.t.
  $\mathrm{Ker}(\mathbf{M}_{r_3}) = \mathrm{Ker}(\mathbf{M}_{r_1}) + \mathrm{Ker}(\mathbf{M}_{r_2})$
- Since:
  - $\dim\left(\mathrm{Ker}(\mathbf{M}_{r_3})\right) \geq \dim\left(\mathrm{Ker}(\mathbf{M}_{r_1})\right)$
  - $\mathbf{M}_{r_3}$ has the same shape as $\mathbf{M}_{r1}$

We know $\mathbf{M}_{r_3}$ exists!

- Set $\mathbf{r}_3 = \mathbf{M}_{r_3}(\mathbf{g}_1 + \mathbf{g}_2)$
- **We are done!** We have $\mathbf{M}_{r_3}\mathbf{x} + \mathbf{r_3} = \mathbf{M}_{r_3}\mathbf{z}$

Today: KG Completion Models

- **What we learned so far:**

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓

Summary of Knowledge Graph

- Link prediction / Graph completion is one of the prominent tasks on knowledge graphs
- Introduce **TransE** / **TransR** / **DistMult** / **ComplEx** models with different embedding space and expressiveness
- **Next:** Reasoning in Knowledge Graphs